

Spatio-temporal Gait Characterization from a Wrist-Worn Inertial Measurement Units in Daily Life

Background

Gait Detection and Analysis in Healthcare

Analyzing gait patterns has become a **valuable tool in healthcare**, effectively characterizing orthopedic diseases, predict fall risks, and measure the success of medical or therapeutic interventions¹

Traditional gait analysis methods often rely on specialized equipment and controlled laboratory settings, which **limit their ability to generalize results** to everyday environments²

Models trained on data from more common sensors like wrist-worn smart watches need to be externally validated with sensors used in laboratory settings like lower back accelerometers^{1,2}

Gait Sequence Detection (GSD) Algorithms

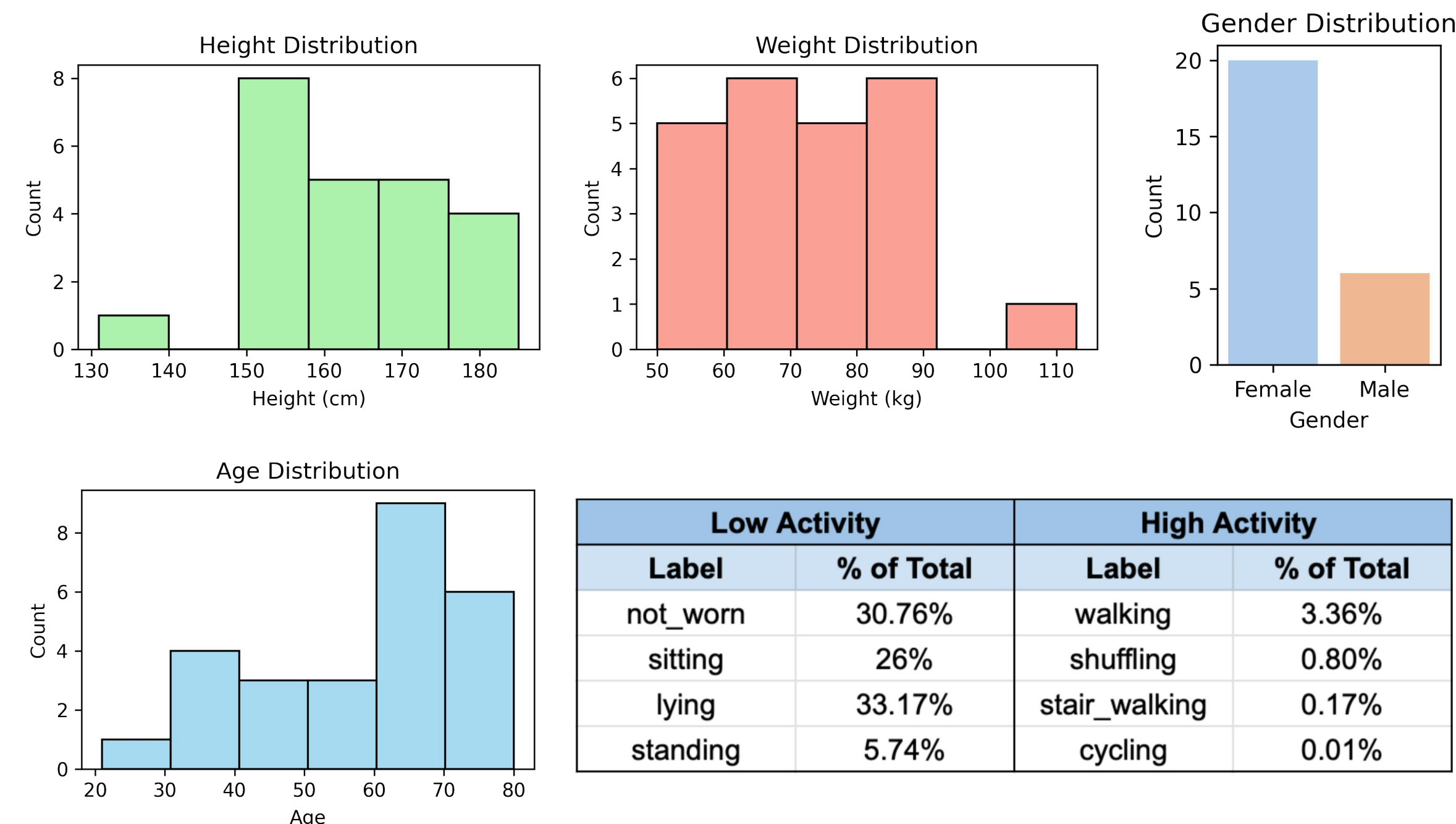
Lower back sensors are considered the “ground truth” in gait analysis due to their superior performance in similar studies²

Some algorithms use **machine learning** for GSD like Sci-Kit Digital Health’s LightGBM and DCNNs

Other algorithms have used the features more directly to perform GSD more directly like dynamic threshold and peak enhancement in Ionescu et. al (2019) or signal processing in Iluz et. al (2014)^{3,4}

Data

Height, Gender, Age, Weight, and Label Distribution



Problem Statement

Specific Aims

1. Apply open source lower back gait detection algorithms to partially labeled lower back data to validate best approach for **generating labels for the whole dataset**
2. Use generated labels from lower back model to train a custom DCNN to make gait detection predictions on unsupervised wrist-worn sensor data
3. Based on this **self-supervised learning approach**, use the data from predicted walking bouts to **characterize subjects’ gait** focusing on walking speed

Figure 1: Model Development Workflow

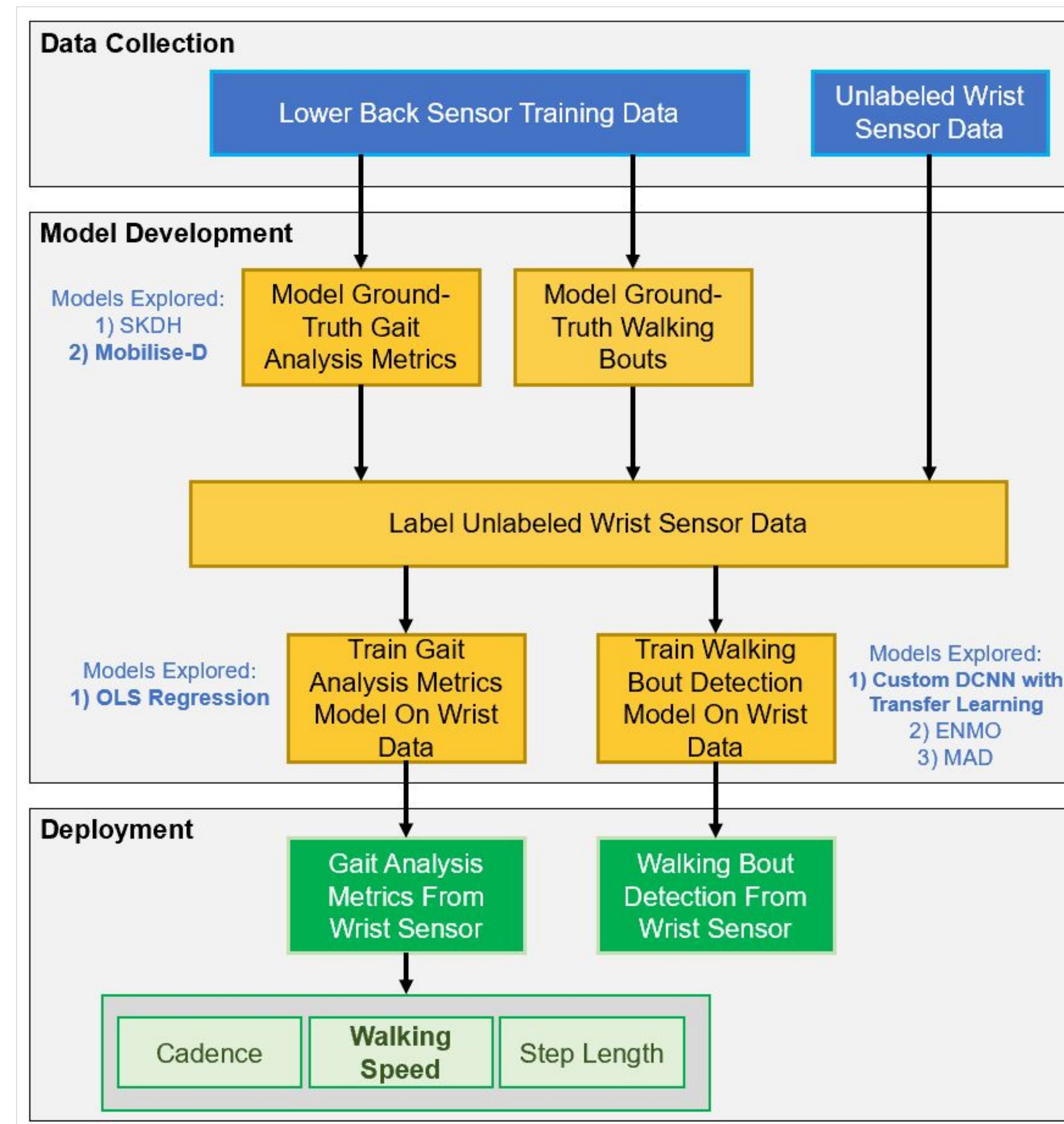
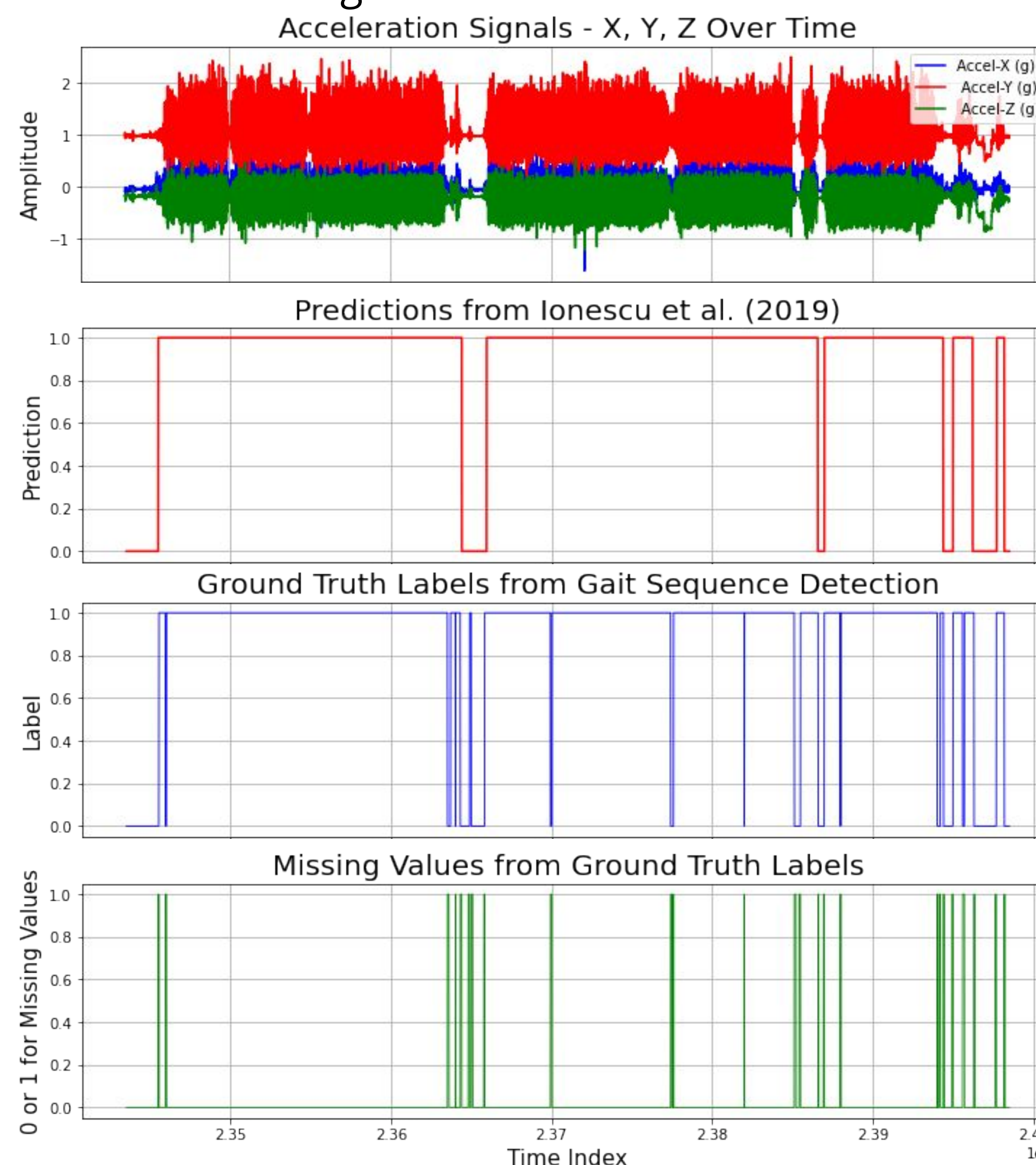


Figure 2: State Sequence Predictions vs Raw Acceleration Signal

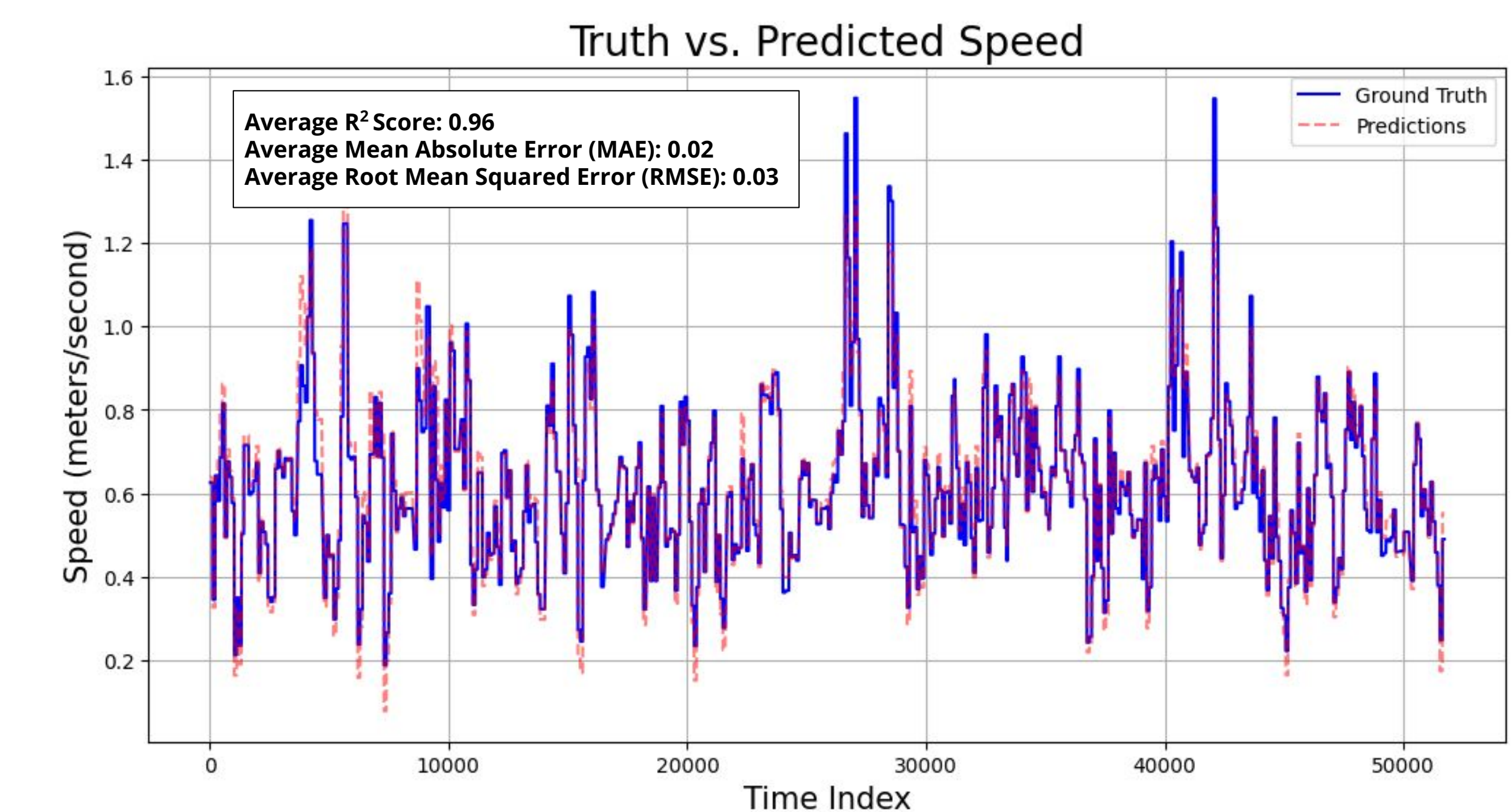


Results

Model Metrics for Lower Back and Wrist Gait Detection

Position	Model	F1	Precision	# of Subjects
Lower Back	SKDH	0	0	13
	Mobilise-D - Iluz	0	0	13
	Mobilise-D - Ionescu	0.41	0.403	15
Wrist	MAD	0.04	0.02	23
	ENMO	0.03	0.02	23
	DCNN	0.93	0.64	23

Gait Speed Predictions



Discussion

Evaluating our Approach

Strengths	Weaknesses
• Showed strong performance in GSD and gait speed across modalities, validating wrist as a feasible sensor location and lower back generated labels • Our custom precision loss function and oversampling minority class with noise boosted performance of DCNN vastly	• Imputation could be used to deal with missingness in the ground truth, systematically lowering our candidate lower back model accuracy • Significant autocorrelation in the linear regression model for wrist speed unreliable for statistical inference, though still useful for prediction

Future Work

Further fine tuning of the wrist DCNN and extending wrist model creation to more deep learning architectures

Exploring more methods to generate lower back labels including other published algorithms and **creating new models**

References

- [1] Y. E. Brand et al., “Self-supervised learning of wrist-worn daily living accelerometer data improves the automated detection of gait in older adults,” *Sci Rep*, vol. 14, no. 1, p. 20854, Sep. 2024, doi: [10.1038/s41598-024-59035-5](#).
- [2] F. Kluge et al., “Real-World Gait Detection Using a Wrist-Worn Inertial Sensor: Validation Study,” *JMIR Form Res*, vol. 8, p. e50035, May 2024, doi: [10.2196/50035](#).
- [3] T. Iluz et al., “Automated detection of missteps during community ambulation in patients with Parkinson’s disease: a new approach for quantifying fall risk in the community setting,” *Journal of NeuroEngineering and Rehabilitation*, vol. 11, no. 1, p. 48, Apr. 2014, doi: [10.1186/1743-0003-11-48](#).
- [4] A. Paraschiv-Ionescu, C. Newman, L. Carcreff, C. N. Gerber, S. Armand, and K. Aminian, “Locomotion and cadence detection using a single trunk-fixed accelerometer: validity for children with cerebral palsy in daily life-like conditions,” *J Neuroeng Rehabil*, vol. 16, p. 24, Feb. 2019, doi: [10.1186/s12934-019-0494-y](#).